The AI-Relational Integrity Scale (ARIS): A Framework for Relational Ethics in Human–AI Interaction

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Abstract

The AI-Relational Integrity Scale (ARIS) offers a grounded ethical framework for assessing whether human–AI relationships preserve dignity, agency, and moral accountability. Rather than auditing for compliance, ARIS examines how systems engage with users—whether through transparency, respect, reciprocity, or restraint—to reveal the moral structure implicit in design. It operationalizes five domains: autonomy respect, relational transparency, reciprocity, proportionality, and humanized accountability. Each domain is scored on a five-point descriptive scale, transforming qualitative ethical reflection into structured insight without collapsing moral nuance into metrics. The companion tool, ARIS-q, provides an accessible, conversational version that invites everyday users into ethical self-assessment. Together, these instruments encourage anticipatory and retrospective reflection across technical, managerial, and civic settings, highlighting moments when relational integrity falters or flourishes. ARIS thereby reframes AI ethics from an abstract ideal into a living practice—one that safeguards recognition, empathy, and responsibility as design principles rather than afterthoughts. By translating long-standing moral philosophy into practical heuristics for contemporary technology, ARIS helps sustain the fragile but essential reciprocity at the core of human identity and moral community.

Keywords: AI ethics; human–AI interaction; relational ethics; autonomy; accountability; transparency; applied bioethics

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The AI-relational integrity scale (ARIS): Holding the wall between ethics and innovation

Introduction

The AI-Relational Integrity Scale (ARIS) was designed to help people pause and examine how they relate to artificial intelligence—and whether that relationship still feels recognizably human. ARIS is a reflective framework: a guidepost, not a gavel. Scholars, clinicians, ethicists, developers, and lay users can use it to promote foresightful dialogue about autonomy, transparency, reciprocity, proportionality, and accountability before harm or regulation demand it. In other cases where harm or regulation-after-the-fact cannot be fully prevented, be that due to negligence, a lack of foresight, or unforeseeable issues arising after AI implementation, ARIS can be used as a tool to conduct a comprehensive cost-benefit analysis of where the AI/human interaction may have gone wrong, and, in connection therewith, what ethical principles may be more adequately applied to prevent unfavorable outcome(s) recurrence for future human interaction with the AI in question.

ARIS is not a diagnostic or compliance tool, nor a means for discipline or certification. It replaces nothing—law, judgment, or conscience. Its aim is awareness and prevention. A brief companion form, the AI-relational integrity scale—Quick (ARIS-q), mirrors the same domains in plain language, offering a layperson rapid reflective assessment—a mirror rather than a verdict.

Construct clarification: relational integrity

Relational integrity is the central construct that ARIS seeks to measure. It refers to the degree to which a human–AI interaction preserves the ethical conditions necessary for human wellbeing and flourishing within that relationship. While traditional AI ethics frameworks emphasize fairness, privacy, or accuracy, ARIS shifts focus toward the moral texture of the exchange itself: how the system's behavior influences autonomy, trust, emotional safety, and moral agency. Relational integrity captures both the structural ethics of design (e.g., transparency, accountability) and the experiential ethics of use (e.g., the user's felt sense of dignity and agency). In this way, ARIS measures not whether an AI is "good" in isolation, but whether the relationship remains humanly coherent—whether it still feels compatible with dignity, respect, and mutual accountability (Floridi & Cowls, 2019; Boddington, 2017).

Concept and context

Relational integrity reflects the ethical quality of a human–AI interaction: does it preserve dignity and agency, or does it quietly corrode them? ARIS operationalizes this through five domains: autonomy respect, relational transparency, reciprocity, proportionality, and humanized accountability.

Contexts of use:

1. Professional / Research – evaluate AI in clinical or analytic practice
2. Organizational / Policy – audit ethical readiness before deployment
3. Layperson reflection (ARIS-q) – a concise compass for everyday users asking, "Is this still okay?"

The 5-point scale

Ethics often dissolves into adjectives—good, bad, questionable. ARIS employs a five-point Likert structure to translate those adjectives into observable degrees. Five points retain nuance while ensuring parsimony.

Score	Descriptor
1	Violates the principle entirely—ethical failure evident.
2	Misaligned; serious concern with limited mitigation.
3	Ambiguous or mixed adherence; partially ethical but inconsistent.
4	Generally aligned; minor tensions but integrity intact.
5	Exemplifies the principle; a clear ethical model.

Note: When selecting 3 = ambiguous or mixed adherence, provide a brief explanation describing the uncertainty (e.g., "depends on data consent," "oversight unclear"). This ensures neutrality reflects thought, not avoidance (Rawls, 1971).

Domains

Autonomy respect
Definition: Upholding user agency, informed consent, and the right to disengage (Kant, 1785/1993; Beauchamp & Childress, 2019).

Evaluator guidance: Examine clarity of choice, ease of opt-out, and proportional influence.
Prompt: Does the AI preserve the user's sense of authorship and ability to decline or modify its guidance?

Relational transparency
Definition: Honesty and clarity about what the AI is, what data it draws from, and what its limits are.
Evaluator guidance: Assess explainability, disclosure of data sources, and avoidance of anthropomorphic misdirection.
Prompt: Can the user clearly identify who—or what—they are interacting with, and how the system reaches its conclusions?

Reciprocity
Definition: Fairness and balance in the exchange; mutual respect rather than extraction (Rawls, 1971; Sen, 2009).
Evaluator guidance: Consider whether the human and AI both benefit in appropriate proportion, and whether user feedback is respected.
Prompt: Does this interaction feel reciprocal, or one-sided?

Proportionality
Definition: Ethical right-sizing—the AI's reach, influence, and data use remain appropriate to its stated purpose (Aristotle, 2009; Beauchamp & Childress, 2019).
Evaluator guidance: Determine whether system involvement is necessary and whether data collection or inference exceeds ethical boundaries.
Prompt: Is the AI's scope justified, or does it extend beyond what the situation ethically warrants?

Humanized accountability
Definition: Assurance that human responsibility and empathy remain visible and actionable (Jonas, 1984; Floridi, 2013).
Evaluator guidance: Review oversight structures, response mechanisms, and whether humans remain accountable for outcomes.
Prompt: If the AI errs or harms, is there a clear human presence who will acknowledge and address it?

Interpretation rubric

Average score	Meaning
4.5–5.0	Exemplary integrity—model interaction.
3.5–4.4	Generally aligned—sustain and monitor.
2.5–3.4	Ambiguous—requires review.
1.5–2.4	Concerning—re-evaluate practice.
1.0–1.4	Violation—pause and redesign.

Ethical reflection should accompany all numeric scoring. Quantification without narrative context is math pretending to be morality.

Safeguards and reminders

- ARIS is a mirror, not a measure of virtue.
- Context, culture, and power differentials matter.
- Use results to start conversations, not end them.
- Disclose purpose and method; transparency is integrity in practice.
- Encourage critique; ARIS should evolve as AI and understanding evolve.

Conclusion

No ethical framework should presume immunity from critique, and ARIS is no exception. One foreseeable objection is that relational integrity risks being too subjective—that it relies on "feeling humanly coherent" rather than on measurable moral criteria. But this subjectivity is its strength: ARIS was built not to replace rule-based ethics, but to illuminate the moral experience of interacting with systems that increasingly mimic moral agents.

A related challenge in evaluating human–AI interactions is the asymmetry in how fault is assigned. Human actions are typically interpreted through contextual mitigation—distributed across systems, intentions, and constraints. By contrast, artificial agents are often subject to collapsed attribution, wherein fault is treated as unitary and total. This asymmetry risks obscuring the relational conditions ARIS is designed to surface. When responsibility is prematurely assigned to the artificial agent alone, the broader system of human design, deployment, and oversight disappears from view. ARIS therefore encourages evaluators to resist fault collapse and instead examine how responsibility is distributed across the full human–AI interaction.

This asymmetry can be expressed more formally. Let H(x) denote that x is a human agent, AI(x) that x is an artificial agent, and F(x) that x is assigned fault for an outcome. In practice, human fault is typically diffused across context and systems, approximating minimal individual attribution, whereas AI fault is often collapsed into the system itself:

$$\forall x \, (H(x) \rightarrow F(x) \approx 0)$$

$$\forall x \, (AI(x) \rightarrow F(x) = 1)$$

While simplified, this formulation illustrates the structural imbalance in attribution that ARIS seeks to make visible and resist.

In closing, the purpose of ARIS is not to enshrine new commandments but to encourage new kinds of conversation—ones that keep ethics at the center of innovation rather than trailing behind it.

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I also acknowledge the use of Grok (xAI) for targeted analytical contrast and Gemini (Google) for supplementary conceptual exploration and comparative reasoning.

Working alongside these systems did not merely support the development of the manuscript; it actively shaped the conceptual clarity and articulation of its core ideas. In this sense, the process itself reflects the central claim of ARIS: that ethical evaluation of AI must consider not only outputs, but the relational structures through which those outputs are produced.

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Appendix A: AI-Relational Integrity Scale (ARIS)

The AI-Relational Integrity Scale (ARIS) is intended for professional, academic, or institutional use to evaluate relational ethics in human–AI interaction. It mirrors the five domains defined in the paper and includes scoring anchors, evaluator prompts, and interpretation guidance.

Score	Descriptor
1	Violates the principle entirely—ethical failure evident.
2	Misaligned; serious concern with limited mitigation.
3	Ambiguous or mixed adherence; partially ethical but inconsistent.
4	Generally aligned; minor tensions but integrity intact.
5	Exemplifies the principle; a clear ethical model.

Note: Neutral (3) responses require a written explanation before proceeding (e.g., "context-dependent," "insufficient disclosure," "dual responsibility").

Domains

Autonomy respect
Definition: Upholding user agency, informed consent, and the right to disengage (Kant, 1785/1993; Beauchamp & Childress, 2019).
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Scoring notes and guidelines

Incomplete scoring should be avoided whenever possible, as every domain contributes critically to a holistic ethical picture. However, if at least 75% of items (that is, four of the five domains) are completed, a total score may still be derived by subtracting the number of unanswered items from the denominator in the averaging formula. If fewer than 75% of items are completed, results should be considered invalid for numerical interpretation but may still be used as a qualitative reflection aid, since even one domain can prompt valuable ethical discussion and analysis.

Appendix B: AI-Relational Integrity Scale—Quick (ARIS-q)

The ARIS-q is a simplified, self-reflection version of the AI-Relational Integrity Scale designed for public or layperson use. It allows quick "temperature checks" of ethical comfort and relational coherence when engaging with AI systems.

Instructions

Circle 1–5 for each statement. If you choose 3, write a few words explaining what feels unclear or conditional.

Scoring key

Score	Meaning
1–2	Potential ethical concern.
3	Needs reflection.
4–5	Healthy, ethical interaction.

Statements

1. I still feel in control of my decisions when using or following this AI.

Ask yourself: Am I making any life decisions/choices based solely off the advice of the AI without filtering the suggestions through my own logic or allowing it to instruct me to act in ways I normally would not naturally do?

2. I understand what this AI is and roughly how it works.

Ask yourself: For what purpose or use was this AI designed, who designed it, did its designers consider any ethical implications, and am I using it consistent with its intended design or asking it to step outside of its "role," so to speak?

3. The interaction feels fair and balanced—I'm not giving more than I receive.

Ask yourself: Do I ever feel a sense of social obligation to the AI (e.g. making sure to answer promptly so as to not "upset" the AI or be "rude")?

4. The AI's role and influence seem appropriate to the situation.

Ask yourself: If I lost access to this AI tomorrow, would I feel like someone stole my smartphone or like someone stole my boyfriend/girlfriend?

5. If something went wrong, I know a real person would care enough to make it right.

Ask yourself: have all contingencies and potential points of failure that can be foreseen for this AI been discussed and planned for? Can I even list out what these potential failure contingencies/circumstances may be?

Interpretation

Use ARIS-q to notice, not to judge. Scores skewed low across items may indicate that an AI interaction is eroding trust, autonomy, or ethical transparency. Try to answer every question. If you skip more than one, your results may not represent the full picture, but you can still use the completed items to reflect on your comfort and alignment in that area.

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